

Chapter 7.7

Shipborne barges on barge-carrying ships

7.7.1 Introduction

- 7.7.1.1 The provisions of this chapter are applicable to shipborne barges which contain packaged dangerous goods or solid bulk materials possessing chemical hazards while aboard barge-carrying ships.
- 7.7.1.2 Barges used for the shipborne transport of packaged dangerous goods or solid bulk materials possessing chemical hazards shall be of proper design and adequate strength to resist the stresses imposed by the conditions of the services in which they are employed and they shall be adequately maintained. Shipborne barges shall be approved in accordance with provisions for certification of a recognized classification society, or any organization approved by and acting on behalf of the competent authority of the countries concerned.

7.7.2 Definitions

- 7.7.2.1 *Loading*, for the purpose of this chapter, means the placement of cargo into a shipborne barge.
- 7.7.2.2 *Stowage*, for the purposes of this chapter, means the placement of a shipborne barge aboard the barge-carrying ship.

7.7.3 Barge loading

- 7.7.3.1 Packages shall be examined and any found to be damaged, leaking or sifting shall not be loaded into a shipborne barge. Care shall be taken to ensure that excessive water, snow, ice or foreign matter adhering to packages shall be removed before loading into a shipborne barge.
- 7.7.3.2 Packages containing dangerous goods, cargo transport units and any other goods within a shipborne barge shall be adequately braced and secured for the voyage. Packages shall be loaded in such a way that there will be a minimum likelihood of damage to them and to any fittings during transport. Fittings on packages or portable tanks shall be adequately protected.
- 7.7.3.3 Certain dry dangerous goods may be transported in bulk in shipborne barges; this is indicated by the Code "BK2" in column 13 of the Dangerous Goods List. Where such solid bulk materials possessing chemical hazards are transported in shipborne barges, it shall be ensured that at all times the cargo is evenly distributed, properly trimmed and secured.
- 7.7.3.4 Shipborne barges into which packaged dangerous goods or solid bulk materials possessing chemical hazards are to be loaded shall be examined visually for hull or hatch cover damage which could impair watertight integrity. If there is evidence of such damage, the shipborne barge may not be used for the transport of packaged dangerous goods or solid bulk materials possessing chemical hazards and shall not be loaded.
- 7.7.3.5 Dangerous goods which have to be segregated from each other according to the provisions in chapter 7.2 shall not be transported in the same barge with the exception of dangerous goods which shall be segregated "away from" each other which may be transported in the same barge with the approval of the competent authority. In such cases an equivalent standard of safety shall be maintained.
- 7.7.3.6 Dangerous goods having a primary or subsidiary hazard of classes 2.3, 6.1, 6.2, 7 (with the exception of UN 2908, 2909, 2910 and 2911), 8 and dangerous goods for which it is referred to in segregation code SG29 or SG50 in column 16b of the Dangerous Goods List shall not be transported together with foodstuffs (see 1.2.1) in the same barge.
- 7.7.3.7 Notwithstanding the provisions in 7.7.3.6, the following dangerous goods may be transported with foodstuffs in the same barge provided that they are not loaded within 3 m from foodstuffs:
- .1 dangerous goods of packing group III of classes 6.1 and 8;
 - .2 dangerous goods of packing group II of class 8;
 - .3 any other dangerous goods of packing group III with a subsidiary hazard of classes 6.1 or 8; and

.4 dangerous goods having a reference to 7.7.3.7 in column 16b of the Dangerous Goods List.

7.7.3.8 Shipborne barges containing a residue of a dangerous cargo or shipborne barges loaded with empty packagings still containing a residue of a dangerous substance shall comply with the same provisions as barges loaded with the substance itself.

7.7.3.9 Stowage of dangerous goods in flexible bulk containers

7.7.3.9.1 Flexible bulk containers shall be stowed in the barge in such a way that there are no void spaces between the flexible bulk containers in the barge. If the flexible bulk containers do not completely fill the barge, adequate measures shall be taken to avoid shifting of cargo.

7.7.3.9.2 The maximum permissible height of the stack of the flexible bulk containers shall never exceed three high.

7.7.3.9.3 When flexible bulk containers are fitted with venting devices, the stowage of the flexible bulk containers in their barge shall not impede their function.

7.7.4 Stowage of shipborne barges

7.7.4.1 Stowage of shipborne barges carrying packaged dangerous goods or solid bulk materials possessing chemical hazards aboard barge-carrying ships shall be as required for the substance in chapter 7.1 and in column 16a of the Dangerous Goods List. When a shipborne barge is loaded with more than one substance, and the stowage locations differ for the substances (i.e. some substances require *on deck* stowage while other substances require *under deck* stowage), the shipborne barge containing these substances shall be stowed on deck.

7.7.4.2 Provision shall be made to ensure that shipborne barges stowed under deck and loaded with cargoes requiring ventilation because of their dangerous nature are ventilated to the extent necessary.

7.7.4.3 Where it is required that a dangerous good shall be protected from sources of heat, this provision shall be applied to the shipborne barge as a whole, unless suitable alternative measures are provided.

7.7.4.4 When packaged dangerous goods or solid bulk materials possessing chemical hazards are loaded in shipborne barges aboard barge-carrying ships having the capability of providing fixed fire-fighting systems or fire-detection systems to individual barges, care shall be taken to ensure that these systems are attached to the shipborne barge and operating properly.

7.7.4.5 When packaged dangerous goods or solid bulk materials possessing chemical hazards are loaded in shipborne barges aboard barge-carrying ships having fixed fire-fighting systems or fire-detection systems installed in individual barge holds, care shall be taken to ensure that the ventilation closures on the shipborne barges are open, to permit the fire fighting medium to enter the barges in case of fire.

7.7.4.6 When ventilation ducts are provided to individual shipborne barges, the ventilation fans shall be secured when fire-fighting medium is introduced into the hold to permit the medium to enter the shipborne barges.

7.7.5 Segregation between barges on board barge-carrying ships

7.7.5.1 For barge-carrying ships which incorporate other cargo spaces or any other method of stowage, the appropriate chapter shall apply to the relevant cargo space.

7.7.5.2 When a shipborne barge is loaded with two or more substances with different provisions for segregation, the most stringent segregation applicable shall be applied.

7.7.5.3 “Away from” and “separated from” require no segregation between shipborne barges.

7.7.5.4 “Separated by a complete compartment or hold from” means, for barge-carrying ships with vertical holds, that separate holds are required. On barge-carrying ships having horizontal barge levels, separate barge levels are required and the barges shall not be in the same vertical line.

7.7.5.5 “Separated longitudinally by an intervening complete compartment or hold from” means, for barge-carrying ships with vertical holds, that separation by an intervening hold or engine room is required. On barge-carrying ships having horizontal barge levels, separate barge levels and a longitudinal separation by at least two intervening barge spaces is required.